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Inventor(s)

Ozanne; Jeffrey A. et al.

Aiming Apparatus

Abstract

In some embodiments, an apparatus comprises a first body and a second body aligned upon an axis. The first body comprises a first sight window and a first side edge, a second side edge and a base edge defining the first sight window, and a first central indicator. The second body comprises a second sight window and a first side edge, a second side edge and a base edge defining the second sight window, and a second central indicator. The axis extends through a center of the first sight window and through a center of the second sight window.

Inventors: Ozanne; Jeffrey A. (La Crosse, WI), Ozanne; Jeffrey J. (Norwalk, WI)

Applicant: Ozanne; Jeffrey A. (La Crosse, WI); Ozanne; Jeffrey J. (Norwalk, WI)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit and is a continuation of U.S. patent application Ser. No. 18/382,382, filed Oct. 20, 2023, which claims the benefit and is a continuation of U.S. patent application Ser. No. 17/314,704, filed May 7, 2021, which claims the benefit of U.S. Patent Application No. 63/021,091, filed May 7, 2020, the entire content of each of which is hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

[0002] This invention relates generally to sighting devices and more particularly to sights used on projectile launching devices, optical devices and other suitable devices used to view a target. Iron sights are generally known in the art as a pair of physical aiming structures located a distance from one another. When positioned and aligned appropriately, these aiming structures help align a device with the target.

[0003] Traditional iron sights use a front sight having a central pin or indicator, which helps to align the device on the center of the target; however, because the pin extends to the center, it generally occludes a portion of the target. A drawback to traditional iron sights is occlusion of the target by the sight structures. There remains a need for novel sighting structures that allow for fast target acquisition, better target visibility and added precision.

[0004] All US patents and applications and all other published documents mentioned anywhere in this application are incorporated herein by reference in their entirety.

[0005] Without limiting the scope of the invention a brief summary of some of the claimed embodiments of the invention is set forth below. Additional details of the summarized embodiments of the invention and/or additional embodiments of the invention may be found in the Detailed Description of the Invention below.

[0006] A brief abstract of the technical disclosure in the specification is provided as well only for the purposes of complying with 37 C.F.R. 1.72. The abstract is not intended to be used for interpreting the scope of the claims.

BRIEF SUMMARY OF THE INVENTION

[0007] In some embodiments, an apparatus comprises a first body and a second body aligned upon an axis. The first body comprises a first sight window and a first side edge, a second side edge and a base edge defining the first sight window. The second body comprises a second sight window and a first side edge, a second side edge and a base edge defining the second sight window. The axis extends through a center of the first sight window and through a center of the second sight window.

[0008] In some embodiments, an apparatus comprises a first body and a second body aligned upon an axis. The first body comprises a first sight window, a first indicator located to a first side of the first sight window and a second indicator located to a second side of the first sight window. The second body comprises a second sight window, a third indicator located to a first side of the second sight window and a fourth indicator located to a second side of the second sight window. The third indicator and the fourth indicator are visible through the first sight window. In some embodiments, a width of the first sight window is greater than a distance between the third indicator and the fourth indicator.

[0009] In some embodiments, a method comprises orienting a device comprising a first body comprising a first sight window and a second body comprising a second sight window. The second sight window is visible through the first sight window. A target is visible through the first sight window and the second sight window.

[0010] These and other embodiments which characterize the invention are pointed out with particularity in the claims annexed hereto and forming a part hereof. However, for a better understanding of the invention, its advantages and objectives obtained by its use, reference can be

made to the drawings which form a further part hereof and the accompanying descriptive matter, in which there are illustrated and described various embodiments of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] A detailed description of the invention is hereafter described with specific reference being made to the drawings.

[0012] FIG. **1** shows an embodiment of a device comprising an aiming apparatus.

[0013] FIG. **2** shows another view of an embodiment of an aiming apparatus.

[0014] FIG. **3** shows another view of an embodiment of an aiming apparatus.

[0015] FIG. **4** shows an embodiment of a first body of an aiming apparatus.

[0016] FIG. **5** shows a user's view of an embodiment of an aiming apparatus.

[0017] FIG. **6** shows another embodiment of an aiming apparatus.

[0018] FIG. **7** shows another embodiment of an aiming apparatus.

[0019] FIG. **8** shows an embodiment of an aiming apparatus and a target.

[0020] FIG. **9** shows a device comprising an aiming apparatus in a first torqued state.

[0021] FIG. **10** shows the device of FIG. **9** in a second torqued state.

[0022] FIG. **11** shows a target and a device comprising an aiming apparatus.

[0023] FIG. **12** shows a device aiming toward a target.

[0024] FIG. **13** shows a device aiming toward a target.

[0025] FIG. **14** shows a side view of a device.

[0026] FIG. **15** is a rear view of the device of FIG. **14**.

[0027] FIG. **16** shows a detail of an embodiment of an aiming apparatus.

[0028] FIG. **17** shows a device another embodiment of an aiming apparatus.

[0029] FIG. **18** shows a detail of an embodiment of a second body.

[0030] FIG. **19** shows an embodiment of a device comprising an optical scope.

[0031] FIG. **20** shows an embodiment of a device comprising a rifle.

[0032] FIG. **21** shows an embodiment of a second body.

DETAILED DESCRIPTION OF THE INVENTION

[0033] While this invention may be embodied in many different forms, there are described in detail herein specific embodiments of the invention. This description is an exemplification of the principles of the invention and is not intended to limit the invention to the particular embodiments illustrated.

[0034] For the purposes of this disclosure, like reference numerals in the figures shall refer to like features unless otherwise indicated.

[0035] FIG. **1** shows an embodiment of a device **38** comprising an embodiment of an aiming apparatus **40**. In some embodiments, the aiming apparatus **40** comprises a first body **50** attached to the device **38** and a second body **60** attached to the device. Desirably, the aiming apparatus **40** is arranged to help a user aim the device **38**, for example by visually aligning the first body **50** and second body **60**.

[0036] In some embodiments, a device **38** comprises a firearm. As shown in FIG. **1**, the device **38** comprises a pistol. In some embodiments, a device **38** comprises a rifle, long gun, shotgun or any other suitable type of firearm. In some embodiments, a device **38** comprises any suitable projectile launching device. In some embodiments, a device **38** comprises an optical device such as a spotting scope.

[0037] In some embodiments, the device **38** comprises a rear end **35** and a front end **36**. In some embodiments, the first body **50** is attached to the device **38** at a location that is closer to the rear end **35** than to the front end **36**. In some embodiments, the second body **60** is attached to the device

38 at a location that is closer to the front end **36** than to the rear end **35**.

[0038] FIG. 2 shows the embodiment of an aiming apparatus **40** shown in FIG. 1. In some embodiments, the first body **50** comprises a sight window **52**. In some embodiments, the sight window **52** comprises an open space. In some embodiments, the sight window **52** comprises an airspace. In some embodiments, the sight window **52** allows visible light to pass through the sight window **52**. In some embodiments, a sight window **52** can have any suitable size, shape and configuration. Desirably, the sight window **52** allows a viewing area or targeting area located in front of the first body **50** to be seen by a user positioned behind the first body **50**.

[0039] In some embodiments, the first body **50** comprises a first raised portion **54** located to a first side of the sight window **52**. In some embodiments, the first body **50** comprises a second raised portion **56** located to a second side of the sight window **52**. In some embodiments, the first raised portion **54** comprises an edge **55** that comprises a first side of the sight window **52**. In some embodiments, the second raised portion **56** comprises an edge **57** that comprises a second side of the sight window **52**. In some embodiments, the first edge **55** and second edge **57** comprise opposed sides of the sight window **52**. In some embodiments, the first body **50** comprises a central portion **46** comprising an edge **47**. In some embodiments, the edge **47** comprises a bottom side or base edge of the sight window **52**.

[0040] In some embodiments, the sight window **52** is generally rectangular. In some embodiments, the first edge **55** is parallel to the second edge **57** and perpendicular to the edge **47** that comprises the bottom of the sight window **52**. In some embodiments, a top edge **58** of the first raised portion **54** is parallel to a top edge **59** of the second raised portion **56**. In some embodiments, the sight window **52** comprises a center **51**.

[0041] In various embodiments, the first sight window **52** can have any suitable shape. Desirably, at least a portion of the second body **60** is viewable through the first sight window **52**. In some embodiments, the first edge **55** and second edge **57** are nonparallel. In some embodiments, the first sight window **52** comprises a trapezoidal shape. In some embodiments, the first edge **55** and second edge **57** comprise curvature.

[0042] In some embodiments, the first body **50** comprises a planar surface **48**. **10** In some embodiments, the planar surface **48** comprises the edges **47**, **55**, **57** that define the sight window **52**.

[0043] In some embodiments, the first body **50** comprises a first indicator **70**. In some embodiments, the first body **50** comprises a second indicator **72**. In some embodiments, the first raised portion **54** comprises the first indicator **70**. In some embodiments, the second raised portion **56** comprises the second indicator **72**. In some embodiments, an indicator **70**, **72** comprises a visual indicator arranged to visually contrast with adjacent portions of the first body **50**. For example, in some embodiments, the planar surface **48** comprises a dark color and the indicators **70**, **72** comprise a light, bright or otherwise contrasting color. In some embodiments, an indicator **70**, **72** comprises a light transmitting or light producing device, such as a light, an LED, a fiber optic line or any other suitable device or material. In some embodiments, an indicator **70**, **72** comprises a radioluminescent material such as tritium.

[0044] An indicator **70**, **72** can comprise any suitable size and shape. In some embodiments, an indicator **70**, **72** is round. In some embodiments, an indicator **70**, **72** is square. In some embodiments, an indicator **70**, **72** is rectangular. In some embodiments, the second indicator **72** comprises a size and shape that is similar to the first indicator **70**. In some embodiments, the second indicator **72** is positioned symmetrically with respect to the first indicator **70** across the sight window **52**. In some embodiments, a distance between the top edge **58** of the first raised portion **54** and the center of the first indicator **70** is similar to a distance between the top edge **59** of the second raised portion **56** and the center of the second indicator **72**. In some embodiments, a distance from the center **51** of the sight window **52** to the center of the first indicator **70** is similar to a distance from the center **51** of the sight window **52** to the center of the second indicator **72**.

[0045] In some embodiments, the first body **50** comprises a central indicator **45** located below the sight window **52**. In some embodiments, the central indicator **45** is positioned on a midline of the first body **50**. In some embodiments, the central indicator **45** comprises a channel extending into the central portion **46** of the first body **50**. In some embodiments, the central indicator **45** interrupts the base edge **47** of the sight window **52**.

[0046] In some embodiments, the first body **50** comprises a first indicator **70** located to a first side of the sight window **52**, a second indicator **72** located to a second side of the sight window **52** and a third indicator **45** located to a third side of the sight window **52**. In some embodiments, the third indicator **45** is located adjacent to the base edge **47** of the sight window **52**. In some embodiments, the first indicator **70** and second indicator **72** are similar to one another. In some embodiments, the third indicator **45** is different from the first indicator **70** and the second indicator **72**.

[0047] In some embodiments, the first body **50** comprises a base portion **53** arranged to engage a device **38**. In some embodiments, the base portion **53** comprises a trapezoidal or dovetail shape arranged to engage a complimentary shaped slot, for example in a device **38**. In some embodiments, a base portion **53** comprises a protrusion arranged to occupy a mounting hole or slot in the device **38**. In some embodiments, a base portion **53** comprises a threaded aperture arranged to engage a fastener.

[0048] In some embodiments, the second body **60** comprises a second sight window **62**. In some embodiments, the second sight window **62** comprises an open space. In some embodiments, the second sight window **62** comprises an airspace. In some embodiments, the second sight window **62** allows visible light to pass through the second sight window **62**. In some embodiments, a second sight window **62** can have any suitable size, shape and configuration. Desirably, the second sight window **62** allows a targeting area located in front of the second body **60** to be seen by a user positioned behind the second body **60**. Desirably, the second sight window **62** excludes any sighting pin or other structure that would occlude a center **61** of the second sight window **62**.

[0049] In some embodiments, the second body **60** comprises a first raised portion **64** located to a first side of the second sight window **62**. In some embodiments, the second body **60** comprises a second raised portion **66** located to a second side of the second sight window **62**. In some embodiments, the first raised portion **64** comprises an edge **65** that comprises a first side of the second sight window **62**. In some embodiments, the second raised portion **66** comprises an edge **67** that comprises a second side of the second sight window **62**. In some embodiments, the first edge **65** and second edge **67** comprise opposed sides of the second sight window **62**. In some embodiments, the second body **50** comprises a central portion **44** comprising an edge **41**. In some embodiments, the edge **41** comprises a bottom side or base edge of the second sight window **62**.

[0050] The second sight window **62** can have any suitable size and shape. In some embodiments, a width of the second sight window **62** is less than a width of the first sight window **52**.

[0051] In some embodiments, the first edge **55** of the first sight window **52** is parallel to the first edge **65** of the second sight window **62**. In some embodiments, the second edge **57** of the first sight window **52** is parallel to the second edge **67** of the second sight window **62**. In some embodiments, the base edge **47** of the first sight window **52** is parallel to the base edge **41** of the second sight window **62**.

[0052] In some embodiments, the second body **60** comprises a planar surface **42**. In some embodiments, the planar surface **42** comprises the edges **41**, **65**, **67** that define the second sight window **62**.

[0053] In some embodiments, the second body **60** comprises a first indicator **74**. In some embodiments, the second body **60** comprises a second indicator **76**. In some embodiments, for example when the first body **50** comprises indicators **70**, **72**, the second body **60** can be considered to comprise a third indicator **74** and a fourth indicator **76**. In some embodiments, the first raised portion **64** comprises the third indicator **74**.

[0054] In some embodiments, the second raised portion **66** comprises the fourth indicator **76**. In

some embodiments, the indicators **74, 76** in the second body **60** can have any suitable size, shape and configuration, for example as described with respect to indicators **70, 72** in the first body **50**. In some embodiments, indicators **74, 76** in the second body **60** are similar to indicators **70, 72** in the first body **50**.

[0055] In some embodiments, a width of the first sight window **52** is greater than a distance between the indicators **74, 76** of the second body **60**, and at least a portion of each indicator **74, 76** is visible through the first sight window **52**. In some embodiments, a width of the first sight window **52** is greater than a distance between the outermost portions of the indicators **74, 76** of the second body **60**, and the entire indicators **74, 76** are visible through the first sight window **52**.

[0056] In some embodiments, the second body **50** comprises a central indicator **43** located below the second sight window **62**. In some embodiments, the central indicator **43** is positioned on a midline of the second body **60**. In some embodiments, the central indicator **43** comprises a channel extending into the central portion **44** of the second body **60**. In some embodiments, the central indicator **43** interrupts the base edge **41** of the second sight window **62**.

[0057] In some embodiments, the second body **60** comprises a first indicator **74** located to a first side of the second sight window **62**, a second indicator **76** located to a second side of the second sight window **62** and a third indicator **43** located to a third side of the second sight window **62**. In some embodiments, the third indicator **43** is located adjacent to the base edge **41** of the second sight window **62**. In some embodiments, the first indicator **74** and second indicator **76** are similar to one another. In some embodiments, the third indicator **43** is different from the first indicator **74** and the second indicator **76**.

[0058] In some embodiments, the second body **60** comprises a base portion **63** arranged to engage a device **38**. In some embodiments, the base portion **63** comprises a protrusion arranged to occupy a mounting hole or slot in the device **38**. In some embodiments, the base portion **63** comprises a threaded aperture arranged to engage a fastener. In some embodiments, the base portion **63** comprises a trapezoidal or dovetail shape arranged to engage a complimentary shaped slot in a device **38**.

[0059] FIG. **3** shows embodiments of a first body **50** and a second body **60** from a viewing position located to the front of the aiming apparatus **40**.

[0060] In some embodiments, the first body **50** comprises one or more upper surfaces **84, 85, 86** oriented at an angle to the axis **80**. In some embodiments, an upper surface **84, 85, 86** is arranged to decline as the surface is traversed in a direction from the shooter toward the target. In some embodiments, the central portion **46** of the first body **50** comprises an upper surface **85** arranged to decline away from the axis **80**. This helps to ensure that the upper surface **85** is not visible to a shooter, and that the base edge **47** defining the sight window **52** appears as a sharp line. In some embodiments, an upper surface **84** of the first raised portion **54** is oriented at an angle to the axis **80**. In some embodiments, an upper surface **86** of the second raised portion **56** is oriented at an angle to the axis **80**. This helps to ensure that the upper surfaces **84, 86** are not visible to the shooter and the top edges **58, 59** of the raised portions **54, 56**, which help to define a top edge of the sight window **52**, appear as sharp lines.

[0061] In some embodiments, the upper surface **84** of the first raised portion **54** is parallel to the upper surface **86** of the second raised portion **56**. In some embodiments, the upper surface **85** of the central portion **46** is parallel to the upper surface **84** of the first raised portion **54**.

[0062] In some embodiments, a channel **45** extends into the upper surface **85** of the central portion **46** of the first body **50**. In some embodiments, a base of the channel **45** extends parallel to the axis **80**.

[0063] In some embodiments, the second body **60** comprises one or more upper surfaces **87, 88, 89** oriented at an angle to the axis **80**. In some embodiments, the upper surfaces **87, 88, 89** of the second body **60** are configured similarly to the upper surfaces **84, 85, 86** of the first body **50**.

[0064] In some embodiments, a channel **43** extends into the upper surface **88** of the central portion

44 of the second body **56**. In some embodiments, a base of the channel **43** extends parallel to the axis **80**.

[0065] FIG. **4** shows another view of an embodiment of a first body **50**. In some embodiments, the first body **50** comprises a hook **92**. In some embodiments, the first body **50** comprises a first hook **92** aligned with a second hook **94**. In some embodiments, the first raised portion **54** comprises the first hook **92** and the second raised portion **56** comprises the second hook **94**. In some embodiments, a hook **92**, **94** comprises a concave portion and defines a cavity. In some embodiments, a hook **92**, **94** can be used to engage an external structure to brace the first body **50** and attached portion of a device **38** against movement. For example, a pistol can comprise a handgrip and a slide comprising the first body **50**. A user can hold a portion of a device **38** that is moveable with respect to the first body **50**, such as the handgrip, and can use a hook **92**, **94** to engage any suitable bracing structure and move the handgrip with respect to the slide by applying a force to the handgrip.

[0066] FIG. **5** shows an example of a shooter's view while aiming a device **38** comprising a first body **50** and a second body **60**. The center **51** of the first sight window **52** is aligned with the center **61** of the second sight window **62**. The second sight window **61** is visible within the first sight window **52**. A target can be viewed through the first sight window **52** and through the second sight window **62**.

[0067] In some embodiments, at least a portion of the indicators **74**, **76** of the second body **60** are visible in the first sight window **52**. In some embodiments, the entirety of each indicator **74**, **76** of the second body **60** is visible through the first sight window **52**. In some embodiments, at least a portion of a central indicator **43** of the second body **60** is visible in the first sight window **52**. In some embodiments, a portion of the central indicator **43** of the second body **60** is visible in the central indicator **45** of the first body **50**.

[0068] In some embodiments, indicators **74**, **76** of the second body **60** and indicators **70**, **72** of the first body **50** are generally horizontally aligned when the center **51** of the first sight window **52** is aligned with the center **61** of the second sight window **62**.

[0069] In some embodiments, the axis **80** extends orthogonal to the first sight window **52** and orthogonal to the second sight window **62**. In some embodiments, the axis **80** extends through a center **51** of the first sight window **52** and through the center **61** of the second sight window **62**. In some embodiments, the axis **80** is orthogonal to a reference axis that extends through the indicators **74**, **76** of the second body **60**. In some embodiments, the axis **80** is orthogonal to a reference axis that extends through the indicators **70**, **72** of the first body **50**. In some embodiments, the axis **80** extends in a reference plane that intersects the first indicator **70** and the second indicator **72**. In some embodiments, the reference plane further intersects the third indicator **74** and the fourth indicator **76**.

[0070] FIG. **6** shows embodiment of a handgun **8** comprising a first body **50** and a second body **60**. In some embodiments, the first body **50** comprises an adjustable sighting structure **10**. In some embodiments, the adjustable sighting structure **10** defines a first adjustable cavity **24**. In some embodiments, the first body **50** comprises a moveable portion and a width of the sight window is adjustable.

[0071] In some embodiments, the second body **60** comprises a second sighting structure **9** defining a second cavity **25**. In some embodiments, a size of the second cavity **25** is adjustable. In some embodiments, the second body **60** comprises a moveable portion and a width of the second sight window is adjustable.

[0072] In some embodiments, the width of the first adjustable cavity **24** is proportionally related to the distance that the first body **50** and the second body **60** are spaced apart. In some embodiments, the first adjustable cavity **24** is proportionally related to the distance that the first body **50** is positioned from the user of the device. In some embodiments, the width of the first adjustable cavity **24** is proportionally related to the distance that the first body **50** is spaced apart from the

second body **60** and the user of the device. In some embodiments, the first adjustable cavity **24** is proportionally related to the width of the second cavity **25**. In some embodiments, the first adjustable cavity **24** is proportionally related to the width of the second cavity **25** and the distance that the first body **50** is spaced apart from the second body **60**.

[0073] In some embodiments, the sight cavity **25** has a width that is proportionally related to the distance that the first body **50** and the second body **60** are spaced apart. In some embodiments, the width of the sight cavity **25** is proportionally related to the distance that the second body **60** is positioned from the user of the device. In some embodiments, the width of the sight cavity **25** is proportionally related to the distance that the first body **50** is spaced apart from the second body **60** and the user of the device.

[0074] FIG. **7** shows a rear view of an embodiment of a device **8** comprising a handgrip **11**, a first sighting structure **12** and a second sighting structure **9**. In some embodiments, the first sighting structure **12** comprises a first luminous element **13** and a second luminous element **18**. In some embodiments, a luminous element **13**, **18** comprises a fiberoptic, a light emitting diode, tritium or any appropriate material or mechanism that provides light, translucency or another suitable visual indication.

[0075] In some embodiments, the second sighting structure **9** comprises a first sighting flange **14** and a second sighting flange **16**. In some embodiments, the first sighting flange **14** and the second sighting flange **16** comprise opposed outer sides of the second sighting structure **9**. In some embodiments, the second sighting structure **9** is arranged such that a user viewing a sighting cavity **23** can see the first sighting flange **14** and the second sighting flange **16** extending outward beyond the first sighting structure **12**. In some embodiments, the first sighting flange **14** and the second sighting flange **16** each comprise an indicator, such as a visual indicator arranged to visually contrast with the first body **50**. In some embodiments, an indicator of the first sighting flange **14** and/or the second sighting flange **16** can comprise any material described with respect to other indicators herein (e.g. indicator **70**).

[0076] FIG. **8** shows an example of sighting alignment with the center **7** of a target **5** visible in the sighting cavity **23**. In some embodiments, the first sighting structure **12** will physically occlude a portion of the second sighting structure **9**. In some embodiments, the first sighting flange **14** and the second sighting flange **16** of the second sighting structure **9** extend laterally outward beyond the first sighting structure **12** and are visible. In some embodiments, both the first sighting flange **14** and the second sighting flange **16** are visible when the device **8** is aligned on target. In some embodiments, indicators of the first sighting flange **14** and the second sighting flange **16** are visible.

[0077] In some embodiments, second sighting structure **9** comprises a sight cavity **25** that is sufficiently wide that sidewall portions defining the sight cavity **25** are not visible through the cavity **24** of the first sighting structure **12**.

[0078] FIG. **9** shows an embodiment of a second sighting structure **29** comprising a narrow sight cavity, wherein both sidewalls that define the sight cavity of the second sighting structure **29** are visible through the sight cavity **27** of the first sighting structure **12**. In some embodiments, when indicators **18**, **13** of the first sighting structure **12** are not aligned with indicators **17**, **26** of the second sighting structure **29**, the device **8** is not aimed at the center **7** of the target **5**. In some embodiments, visibility of a top portion of the second sighting structure **29** above the top of the first sighting structure **12** indicates that the device **8** is not aimed at the center **7** of the target **5**.

[0079] FIG. **10** shows a device **8** in a laterally torqued state, wherein the device **8** is not aligned on the center **7** of the target **5**. The second sighting flange **16** is visible but the first sighting flange **14** is not visible. Only one indicator **17** of the second sighting structure **9** is visible.

[0080] FIG. **11** shows a device **8** and a target **5** spaced apart at a distance from one another.

[0081] FIG. **12** shows a device **8** in a vertically torqued state such that the device **8** is not aimed directly at the center **7** of the target **5**. In some embodiments, a first reference axis **28** extends

through indicators **13**, **18** of the first sighting structure **12**. In some embodiments, a second reference axis **30** extends through indicators **15**, **17** of the second sighting structure **29**. In some embodiments, when the first reference axis **28** appears offset from the second reference axis **30**, the device **8** is torqued in a vertical direction. In some embodiments, the first reference axis **28** and the second reference axis **30** will converge when the device **8** is aimed at the center **7** of the target **5**. In some embodiments, the reference axes **28**, **30** extend through a center of each associated indicator. [0082] FIG. **13** shows another potential view while aiming a device **8**. FIG. **14** shows a side view of a device **8** comprising a first sighting structure **10** comprising an adjustment mechanism **19**. FIGS. **15** and **16** show another potential view while aiming the device of FIG. **14**.

[0083] Referring to FIGS. **13** and **16**, in some embodiments, a top surface of the first body **50**/first sighting structure **10** and a top surface of the second body **60**/second sighting structure **9** are aligned with one another. In some embodiments, indicators of the first body **50**/first sighting structure **10** and indicators of the second body **60**/second sighting structure **9** are aligned on a plane that intersects the centers of the sight windows.

[0084] FIGS. **17** and **18** show an embodiment of an adjustable sight structure **10**. In some embodiments, a first body **50** comprises an adjustable sight window **52**. In some embodiments, a width of the sight window **52** is adjustable. In some embodiments, the first body **50** comprises a base **53**, a first portion **75** moveable with respect to the base **53** and a second portion **77** moveable with respect to the base **53**. In some embodiments, the first portion **75** comprises the first raised portion **54**. In some embodiments, the second portion **77** comprises the second raised portion **56**. In some embodiments, the first body **50** comprises a locking mechanism **33** arranged to lock a moveable portion **75**, **77** against movement. In some embodiments, a locking mechanism comprises a fastener or set screw.

[0085] In some embodiments, the first body **50** comprises an adjustment mechanism **19** arranged to move at least one portion **75**, **77** of the first body **50** with respect to the base **53**. In some embodiments, an adjustment mechanism **19** is arranged to move the first portion **75** and the second portion **77** simultaneously with respect to the base **53**. In some embodiments, the adjustment mechanism **19** is arranged to move the first portion **75** and the second portion **77** in equal but opposite directions with respect to the base **53**. In some embodiments, movement of the first portion **75** and second portion **77** is symmetrical across the center of the sight window **52**. In some embodiments, the adjustment mechanism is engaged with a central portion of the base **53**. In some embodiments, the adjustment mechanism **19** comprises a trapezoidal shape. In some embodiments, the adjustment mechanism comprises a wedge. In some embodiments, the base **53** comprises a raised flange, recessed channel or other engagement mechanism and the first portion **75** and second portion **77** are shaped to engage the base **53**. In some embodiments, the first portion **75** and second portion **77** are slidably engaged with the base **53**.

[0086] In some embodiments, the second body **60** comprises a base and one or more portions moveable with respect to the base, for example to adjust a width of the second sight window **62**. In some embodiments, the second body **60** comprises an adjustment mechanism **19** as described herein with respect to the first body **50**.

[0087] FIG. **19** shows an embodiment of a first body **50** and a second body **60** attached to a device **38** comprising an optical scope.

[0088] FIG. **20** shows an embodiment of a first body **50** and a second body **60** attached to a device **38** comprising a rifle. FIG. **21** shows an embodiment of a first body **50** arranged to engage a picatinny rail. In some embodiments, the first body **50** comprises a base **53** configured to engage a picatinny rail. In some embodiments, the first body **50** comprises a first engaging structure **81** and a second engaging structure **83** arranged to engage a mounting structure such as a picatinny rail. In some embodiments, the engaging structures **81**, **83** define a cavity **82** arranged to accept a mounting structure such as a picatinny rail. In some other embodiments, a base **53** can be configured for any suitable known firearm accessory mounting method or rail integration system,

such as a Modular Lock system, Weaver rail, KeyMod, etc.

[0089] Various embodiments of an aiming apparatus **40** comprise sight windows **52**, **62** having any suitable sizing relative to one another. In some embodiments, the first sight window **52** is wider than the second sight window **62**. In some embodiments, the second sight window **62** is wider than the first sight window **52**. In some embodiments, two variables can help to determine proper sight window sizing—a first distance between the first sight window **52** and the second sight window **62**, and a second distance between the first sight window **52** and the user's viewpoint. In some embodiments, the first distance is relatively small compared to the second distance—for example when the device **8** is a pistol. For example, the first distance between the first sight window **52** and the second sight window **62** in a pistol may be in the range of 4"-6" and the second distance is likely in the range of 10"-18". When the second distance is larger than the first distance, it is often desirable to have the first sight window **52** wider than the second sight window **62**. In some embodiments, the first distance is relatively large compared to the second distance—for example when the device **8** is a rifle. For example, the first distance between the first sight window **52** and the second sight window **62** in a rifle may be in the range of 8"-18" and the second distance is likely in the range of 3"-6". When the first distance is larger than the second distance, it is often desirable to have the second sight window **62** wider than the first sight window **52**.

[0090] In some embodiments, a method comprises various steps of providing and/or using the first body **50** and second body **60** as described herein. In some embodiments, a method comprises viewing a target through a first sight window and a second sight window. In some embodiments, a method comprises adjusting a width of a sight window.

[0091] The above disclosure is intended to be illustrative and not exhaustive. This description will suggest many variations and alternatives to one of ordinary skill in this field of art. All these alternatives and variations are intended to be included within the scope of the claims where the term "comprising" means "including, but not limited to." Those familiar with the art may recognize other equivalents to the specific embodiments described herein which equivalents are also intended to be encompassed by the claims.

[0092] Further, the particular features presented in the dependent claims can be combined with each other in other manners within the scope of the invention such that the invention should be recognized as also specifically directed to other embodiments having any other possible combination of the features of the dependent claims. For instance, for purposes of claim publication, any dependent claim which follows should be taken as alternatively written in a multiple dependent form from all prior claims which possess all antecedents referenced in such dependent claim if such multiple dependent format is an accepted format within the jurisdiction (e.g. each claim depending directly from claim **1** should be alternatively taken as depending from all previous claims). In jurisdictions where multiple dependent claim formats are restricted, the following dependent claims should each be also taken as alternatively written in each singly dependent claim format which creates a dependency from a prior antecedent-possessing claim other than the specific claim listed in such dependent claim below.

[0093] This completes the description of the preferred and alternate embodiments of the invention. Those skilled in the art may recognize other equivalents to the specific embodiment described herein which equivalents are intended to be encompassed by the claims attached hereto.

Claims

1. An apparatus comprising: a shooting device comprising a muzzle, a front and a rear, the muzzle located at the front; a first body and a second body attached to the device, the first body and the second body aligned upon an axis; the first body comprising a first sight window, the first body comprising a first side edge, a second side edge and a base edge defining the first sight window, the first body comprising a first central indicator comprising a first channel extending into the base

- edge, the first channel centered across a width of the first sight window; the second body comprising a second sight window, the second body comprising a first side edge, a second side edge and a base edge defining the second sight window, the second body comprising a second central indicator comprising a second channel extending into the base edge, the second channel centered across a width of the second sight window; the axis extending through a center of the first sight window and a center of the second sight window.
2. The apparatus of claim 1, wherein the width of the first sight window is different from the width of the second sight window.
 3. The apparatus of claim 1, wherein a width of the first channel is the same as a width of the second channel.
 4. The apparatus of claim 1, wherein a width of the first channel is less than $\frac{1}{3}$ of the width of the first sight window.
 5. The apparatus of claim 4, wherein a width of the second channel is less than $\frac{1}{3}$ of the width of the second sight window
 6. The apparatus of claim 1, wherein a ratio between a width of the first channel and the width of the first sight window is different from a ratio between a width of the second channel and the width of the second sight window.
 7. The apparatus of claim 1, the first side edge of the first sight window parallel to the first side edge of the second sight window.
 8. The apparatus of claim 1, the first body comprising a top surface oriented non-parallel to the axis.
 9. The apparatus of claim 1, the second body comprising a first indicator and a second indicator, the first indicator located to a first side of the second sight window, the second indicator located to a second side of the second sight window.
 10. The apparatus of claim 9, the first body comprising a third indicator and a fourth indicator, the third indicator located to a first side of the first sight window, the fourth indicator located to a second side of the first sight window.
 11. The apparatus of claim 9, the first indicator comprising a radioluminescent material.
 12. The apparatus of claim 1, the first body comprising an adjustment mechanism arranged to change a width of the first sight window.
 13. The apparatus of claim 1, the first body comprising a hook.
 14. An apparatus comprising: a shooting device comprising a muzzle, a front and a rear, the muzzle located at the front; a first body and a second body attached to the device, the first body and the second body aligned upon an axis; the first body comprising a first sight window, a first indicator located to a first side of the first sight window, a second indicator located to a second side of the first sight window, and a first central indicator comprising a first channel extending into a base edge of the first sight window, the first channel centered across a width of the first sight window; the second body comprising a second sight window, a third indicator located to a first side of the second sight window, a fourth indicator located to a second side of the second sight window, and a second central indicator comprising a second channel extending into a base edge of the second sight window, the second channel centered across a width of the second sight window; wherein the third indicator, the fourth indicator and the second channel are visible through the first sight window.
 15. The apparatus of claim 14, wherein the first indicator, the second indicator, the third indicator, the fourth indicator are aligned on a common plane.
 16. The apparatus of claim 15, wherein a center of the first sight window and a center of the second sight window are aligned on the common plane.
 17. The apparatus of claim 14, wherein the width of the first sight window is different from the width of the second sight window.
 18. The apparatus of claim 17, wherein a width of the first channel is the same as a width of the

second channel.

19. The apparatus of claim 14, wherein a width of the first channel is less than $\frac{1}{3}$ of the width of the first sight window.

20. A method comprising: orienting a device comprising a first body comprising a first sight window comprising a first central indicator comprising a first channel extending into a base edge of the first sight window and a second body comprising a second sight window comprising a second central indicator comprising a second channel extending into a base edge of the second sight window such that the second sight window is visible through the first sight window and a target is visible through the first sight window and the second sight window, wherein the first central indicator is aligned with the second central indicator.
